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Acknowledged Fast, Fixed Slow: Deploying Self-Testing Smoke Alarms Across a Rental Portfolio, and the Fault a Managing Agent Dismisses Quickest but Dispatches Last

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Abstract. Rental-portfolio smoke alarms are typically checked once a year, by an inspector, on a schedule the alarm's own condition has no say in. This paper reports a sixteen-month deployment of self-testing, app-linked interconnected smoke alarms across 214 rental properties, comparing the managing agent's acknowledgment of each weekly self-test fault against how long it actually takes a technician to attend. Acknowledgment is fast and close to uniform across fault types (91.7-95.1% within forty-eight hours), but dispatch within fourteen days ranges from 87.8% for a full alarm failure to 19.1% for the portfolio's most common fault, a low-battery flag, with median attendance lag stretching from four days to sixty-one. Over the same window, the portfolio's unchanged annual paper compliance certificate recorded non-compliance in only 8.9% of properties the new self-test log had already flagged. A pre-registered ablation testing whether a repeated SMS escalation closes the low-battery gap finds it does not: dispatch-by-day-21 moves by 1.8 percentage points ($p = .71$). Acknowledging a fault and acting on it, we suggest, are governed by different institutional processes, and the reminder channel this deployment could most easily add turns out not to be the one the gap runs through.

Introduction

A rental property's smoke alarm is usually checked once: an inspector visits once a year, tests the alarm, signs a certificate, and leaves. Between visits, whether the alarm still functions is not observed by anyone in a position to act on it. Self-testing, app-linked alarms promise to close that gap by reporting a weekly pass or fault directly to whoever manages the property, but a report reaching a managing agent's dashboard is not the same event as a technician reaching the property. The distance between the two — how a fault is acknowledged versus how it is acted on — is the object of this paper.

Continuous-monitoring literatures elsewhere document exactly this distance without necessarily naming it: clinical alert systems distinguish an alert's volume from its clinical value [1], [2], and the wider human-automation literature treats an operator's actual reliance on a signal as governed by different forces than the signal's own reliability [3]. Whether a managing agent's response to a smoke alarm's own fault log shows the same separation, at rental-housing scale, has not to our knowledge been measured against the sixteen-month deployment this paper reports.

This paper instruments a mid-sized rental-property portfolio's self-testing smoke alarm rollout end to end — from the weekly self-test cycle through to the technician's attendance — and tests one intervention aimed at closing the gap the instrumentation surfaces. Its contributions:

1. We report, across 214 rental properties and sixteen months, what we believe is among the first continuously logged comparisons of

a self-testing smoke alarm's fault log against both the managing agent's acknowledgment of that log and the property's unchanged annual paper compliance certificate.

2. We find acknowledgment speed is high and close to uniform across three fault types, while dispatch speed diverges sharply between them, suggesting the two outcomes are governed by separate institutional processes rather than a single triage judgement applied consistently to the log.
3. We test whether a repeated escalation reminder narrows the widest of those gaps and find, on a pre-registered day-21 window, that it does not appear to, which we read as evidence about where the intervention should look next rather than evidence the gap cannot be closed.

Related work

Clinical alarm environments are the most heavily documented setting for the acknowledgment-remediation gap this paper studies elsewhere: monitor alarms generate volumes of low-value alerts that consume attention without changing care [1], and overridden-alert rates in computerised physician order entry vary widely by alert type rather than tracking any single fatigue mechanism [2]. The human-automation literature frames this more generally as a use-misuse-disuse-abuse continuum, in which an operator's reliance on an aid tracks its perceived rather than actual reliability [3]; false traffic-conflict alerts in air-traffic control, for instance, do not straightforwardly desensitise controllers the way a naive account of alarm fatigue would predict [4], evidence that acknowledgment and remediation of the same alert can separate for reasons other than simple habituation.

Smoke alarms carry a substantial randomised-trial literature of their own, concentrated on whether an alarm is present and working rather than on what happens once a fault is logged: giveaway and installation programmes raise ownership without guaranteeing an alarm still functions at follow-up [5], [6], alarm chemistry and battery type shape functional lifespan [7], and targeted health education measurably raises operability [8]. Outside the household, smart-home devices more broadly are shown to be abandoned outright once their signal comes to be seen as irrelevant to the resident's own concerns [9]; this paper's managing agents do not abandon the signal, but the gap between acknowledging it and acting on it amounts to a milder version of the same disengagement. Verified-response policies filtering which security-alarm activations reach a human at all are shown to change downstream police performance without changing the alarm hardware itself [10], and predictive-maintenance alerting is an active engineering literature in its own right, addressing which fault signals are worth raising rather than what happens once one is [11], [12]. Municipal rental-housing inspection research finds that pairing code enforcement with a social-service response, rather than enforcement alone, is what actually moves compliance outcomes [13] — a pairing this paper's own escalation ablation echoes at device scale, without matching its result.

Prior work at this institution has linked a public-pool lifeguard's own vigilance to how long they have worked under an automated drowning-detection alarm, even as the venue's contracted alarm-response-time compliance holds at ceiling throughout [14], and separately deployed a load-cell sensor at a library return chute whose alert log came to be read as evidence of something the sensor was never built to measure [15]. This paper asks a structurally similar question — what a fault log's own acknowledgment conceals — of a rental portfolio's smoke alarms rather than a pool or a library chute.

Method

Deployment

Interconnected, self-testing smoke alarms were retrofitted across 214 properties in a single rental-property management portfolio (685 alarms; every bedroom and connecting hallway, per standard interconnection practice). Each alarm runs a weekly self-test cycle — battery voltage, optical chamber reading against its own installation baseline, and a communication ping to the portfolio's management app — and reports a pass or fault to a property-level dashboard the managing agent reviews. The deployment ran for sixteen months, logging 14,206 property-weeks (vacant weeks excluded).

Each self-test cycle is classified against three fixed thresholds, set before deployment and applied without exception:

```
if not self_test.responded:
FULL_FAILURE
elif abs(chamber - baseline) > drift_τ:
SENSOR_DRIFT
elif battery_voltage < 2.6V:
LOW_BATTERY
else:
PASS
```

Acknowledgment and dispatch

A fault counts as **acknowledged** once the managing agent dismisses its dashboard flag, and as **dispatched** once a technician work order is raised and attended:

$$\text{Acknowledgment rate} = \frac{N_{\text{ack} \leq 48h}}{N_{\text{total}}} \times 100$$

Dispatch lag L_i is measured, for each dispatched fault i, as the number of days between the fault first appearing on the dashboard and the technician's attendance:

$$L_i = d_i^{\text{dispatch}} - d_i^{\text{fault}}$$

Certificate baseline

The portfolio's existing annual paper compliance certificate — an inspector's visit, unmodified by this deployment — continued throughout the study window. For the subset of properties with at least one self-test fault logged, we recorded whether that property's next certificate also recorded non-compliance.

Escalation ablation

Low-battery faults, the deployment's largest class, were split by property-ID parity into two arms, assigned before the fault data were examined: an **escalation** arm receiving a repeated SMS reminder to the managing agent on day 7 if the fault remained undispatched, and a **standard** arm receiving only the single push notification every fault triggers. The day-21 dispatch window was fixed before deployment.

Results

Properties in portfolio	214
Alarms deployed	685
Deployment window	16 months
Property-weeks logged	14,206
Fault events logged	386

Table 1: Deployment parameters.

Of 386 fault events (Table 1), 278 (72.0%) were low-battery flags, 67 (17.4%) sensor drift, and 41 (10.6%) full failures. Acknowledgment within forty-eight hours holds high and close to uniform across all three — 91.7%, 92.5%, and 95.1% respectively — but dispatch within fourteen days does not (Figure 1): 87.8% for full failures, 38.8% for sensor drift, and 19.1% for the far more common low-battery flag.

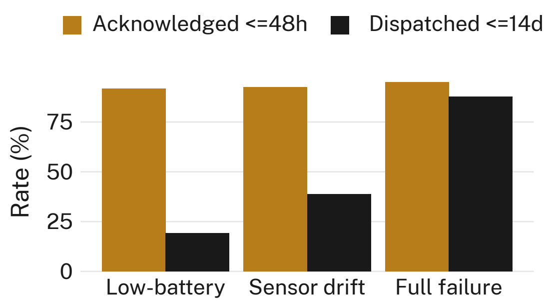

Figure 1: Acknowledgment-within-48h and dispatch-within-14-days rates by fault type. Acknowledgment is flat across fault types; dispatch is not.

Among dispatched faults, the same separation shows in how long attendance actually takes (Figure 2): a median of four days for full failures (IQR 2–7), nineteen days for sensor drift (IQR 11–33), and sixty-one days for low-battery flags (IQR 38–104) — the fault acknowledged fastest of the three waits longest for a technician.

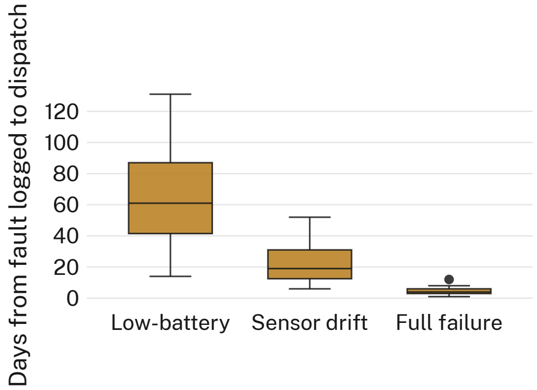

Figure 2: Dispatch lag in days, dispatched faults only, by fault type. Medians separate sharply even though acknowledgment does not.

Of the 158 properties with at least one self-test fault logged during the window, the portfolio's unchanged annual paper certificate recorded non-compliance at only 14 (8.9%) on its next visit — consistent with faults that self-resolve, or recur, between the certificate's once-a-year snapshot and whatever the self-test log has already caught in between.

The escalation ablation, targeting low-battery faults specifically, changes almost nothing (Figure 3): dispatch by day 21 reached 21.4% in the escalation arm (30/140) against 19.6% in the standard arm (27/138), a difference of 1.8 percentage points that falls well short of significance ($\chi^2(1, N = 278) = 0.14, p = .71$). We retain the escalation message in the live deployment for completeness; it does not appear to be the lever this gap runs through.

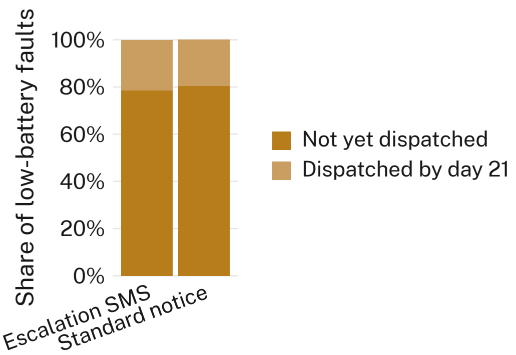

Figure 3: Dispatched-by-day-21 share for low-battery faults, escalation SMS arm ($n = 140$) against standard single-notification arm ($n = 138$). The two arms are barely distinguishable.

Discussion

Acknowledgment and dispatch of the same fault log separate cleanly in this deployment, and the separation is not explained by the low-battery fault being harder to notice — it is acknowledged, if anything, slightly faster than a full failure is. What differs is what happens after acknowledgment, a pattern consistent with reliance and disuse findings from other continuous-monitoring settings [1], [3] rather than with a single triage rule applied evenly across fault types. The certificate comparison suggests the self-test log is, on its own terms, a genuine diagnostic improvement on the annual visit it supplements; the deployment's dispatch figures suggest that improvement is not yet reaching the point where a technician stands in the hallway.

The escalation ablation's null result narrows rather than closes the explanation space. A repeated reminder assumes the managing agent's priority judgement about a low-battery flag would shift given a second nudge on the same channel; the data are more consistent with that judgement being set by the fault's label at first sight, and untouched by how many times the same label repeats. If so, the more promising lever is not reminder frequency but how a low-battery fault is described to the agent in the first place — a question this deployment was not designed to test.

Limitations

This deployment covers one property-management portfolio over sixteen months; a longer window might narrow the acknowledgment-dispatch gap this study's certificate comparison could not observe directly, though the consistency of the pattern across three structurally different fault types suggests it is not confined to one contractor relationship. The escalation ablation's confidence interval on the arm difference (-5.8 to +9.4 percentage points) rules out only an effect larger than roughly nine points, a bound that leaves open a smaller true effect this deployment was not powered to detect. Fault classification relied on the manufacturer's fixed thresholds rather than an independent bench test of each alarm, though the same thresholds were applied identically across all 214 properties throughout.

Conclusion

A self-testing smoke alarm's fault log is acknowledged quickly and almost uniformly across fault types; how long it takes a technician to actually attend is not uniform at all, and the gap is largest for the fault type the portfolio logs most often. A repeated reminder aimed at that gap moved dispatch by less than two percentage points over three weeks. The deployment's own annual paper certificate, unmoved by any of this, continues to record the portfolio as compliant on a schedule the self-test log has already shown to be too coarse to catch what it catches. Future work might test whether relabelling a low-battery fault by its estimated occupant risk, rather than its sensor channel, moves dispatch time where a repeated reminder did not.

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